

“E-Commerce Analytics Project Summary”

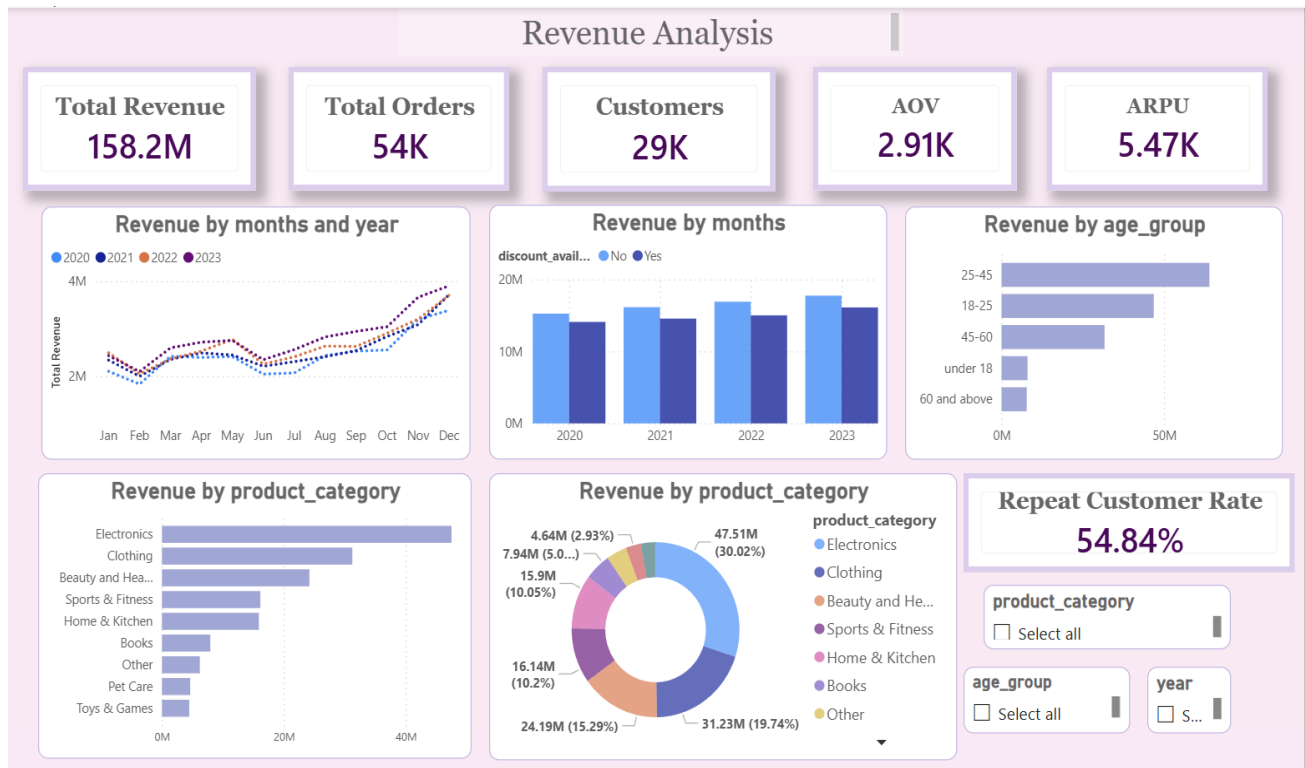
Introduction:

This project focuses on exploratory data analysis and business analytics for an e-commerce dataset.

The main goal was to analyze customer behavior, revenue trends, product performance, and customer retention in order to identify opportunities for business growth and optimization.

The project includes:

- Data cleaning and preprocessing in Python,
- SQL-based analytical queries,
- Exploratory data analysis (EDA),
- RFM customer segmentation,
- Cohort and retention analysis,
- Interactive dashboard in Power BI.



Key Questions

The project aimed to answer the following business questions:

- How does revenue change over time?
- Which customer segments generate the highest revenue?
- How do discounts influence sales performance?
- What are the main purchasing trends by category and year?
- What is the customer retention rate?
- Which customers are the most valuable according to RFM analysis?
- How does Average Order Value (AOV) change over time?
- Which categories and customer groups contribute the most to total revenue?

Dataset Description

The Dataset used for this project spans over a period of 2019 and 2024 years. There are columns such as :

- cid (Customer ID): A unique identifier for each customer.
- tid (Transaction ID): A unique identifier for each transaction.
- gender: The gender of the customer, categorized as Male or Female.
- age_group: Age group of the customer, divided into several ranges.
- purchase-date: The timestamp of when the transaction took place.
- product_category: The category of the product purchased, such as Electronics, Apparel, etc.
- discount_availed: Indicates whether the customer availed any discount (Yes/No).
- discount_name: Name of the discount applied (e.g., FESTIVE50).
- discount_amount_(inr): The amount of discount availed by the customer.
- gross_amount: The total amount before applying any discount.
- net_amount: The final amount after applying the discount.
- purchase_method: The payment method used (e.g., Credit Card, Debit Card, etc.).
- location: The city where the purchase took place.

Main preprocessing steps included:

- handling missing values,
- checking duplicates,
- converting date columns to datetime format,
- cleaning column names,
- detecting negative and abnormal values,
- feature engineering for analytics.

```
data.columns = (  
    data.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(' ', '_'))
```

```
missing_data = data.isnull().sum()
missing_percent = (missing_data/len(data))*100
missing_percent
```

```
data['purchase_date'] = pd.to_datetime(data['purchase_date'],dayfirst=True)
```

```
data.duplicated().sum()
```

Key Metrics Calculated

The following business metrics were calculated during the analysis:

- Total Revenue
- Average Order Value (AOV)
- Average Revenue per User (ARPU)
- Revenue Growth Rate
- Number of Orders
- Number of Unique Customers
- Customer Retention Rate
- Repeat Purchase Rate
- RFM Scores
- Revenue by Category
- Revenue by Year and Month
- Customer Segmentation Metrics

Example formulas:

$\text{AOV} = \text{Total Revenue} / \text{Number of Orders}$

$\text{Retention Rate} = \text{Returning Customers} / \text{Total Customers}$

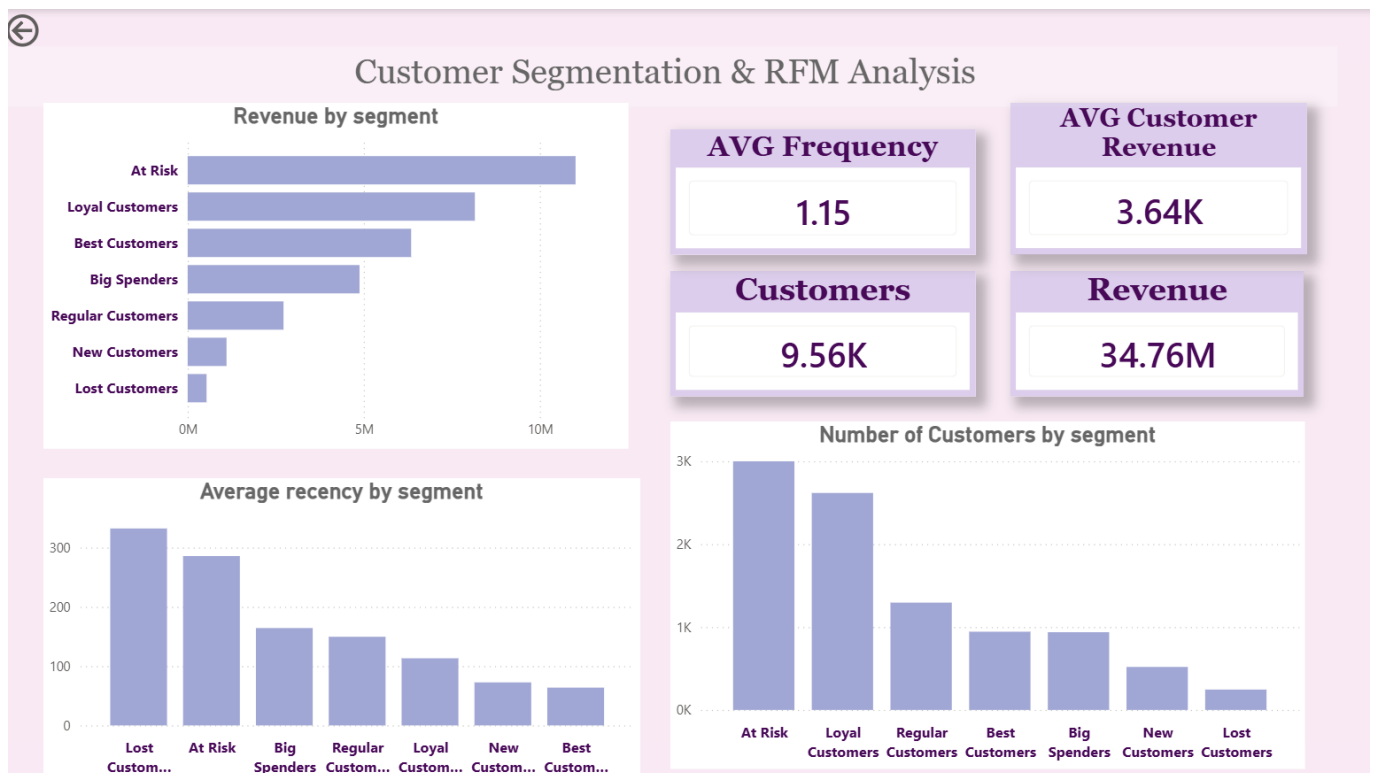
$\text{ARPU} = \text{Revenue} / \text{Number of Users}$

Derived Variables

Several additional analytical variables were created to improve the analysis:

- Order Year
- Order Month
- Customer Age Groups
- Revenue Categories
- Discount Groups
- RFM Scores:
- Recency
- Frequency
- Monetary
- Cohort Month
- Customer Lifetime Indicators
- Repeat Customer Flag

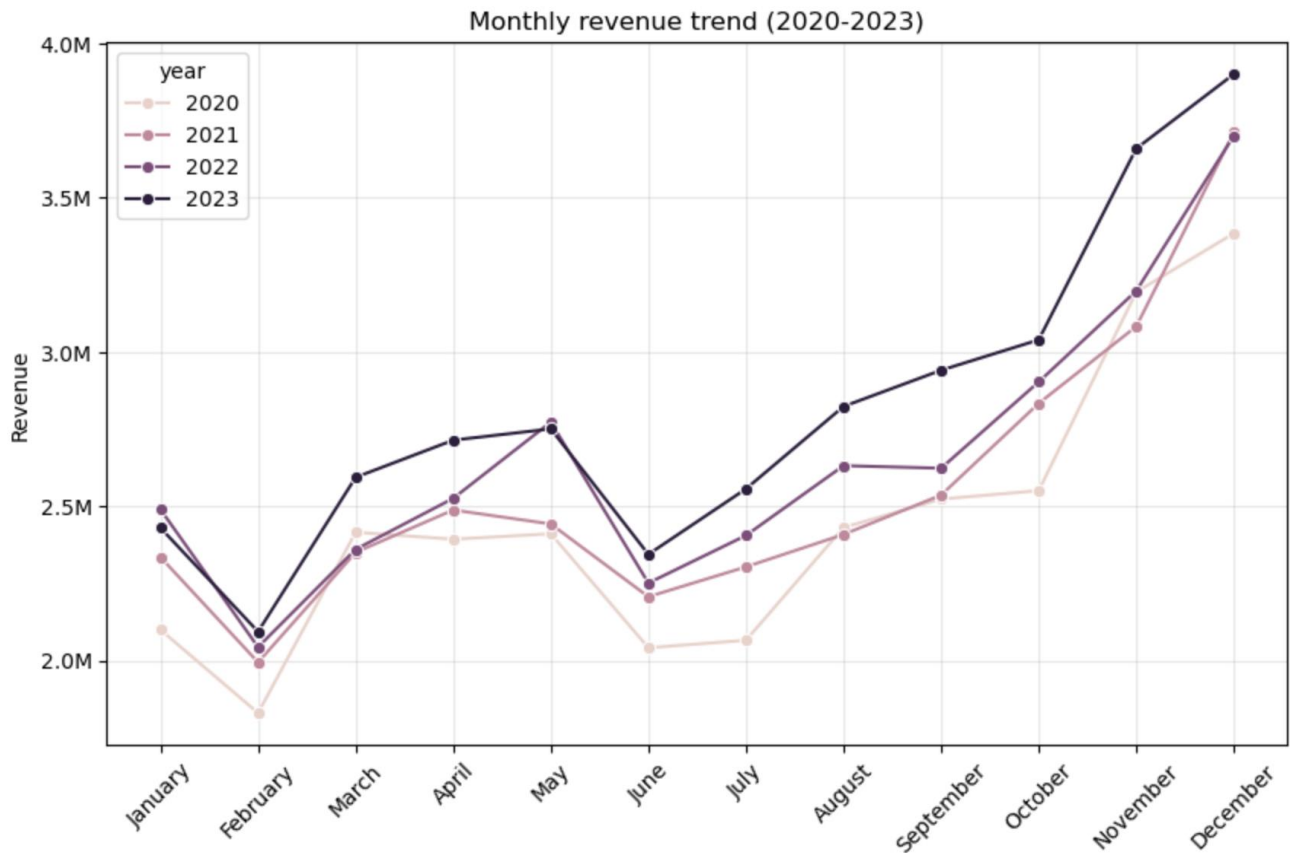
These derived variables helped perform segmentation and trend analysis more effectively.



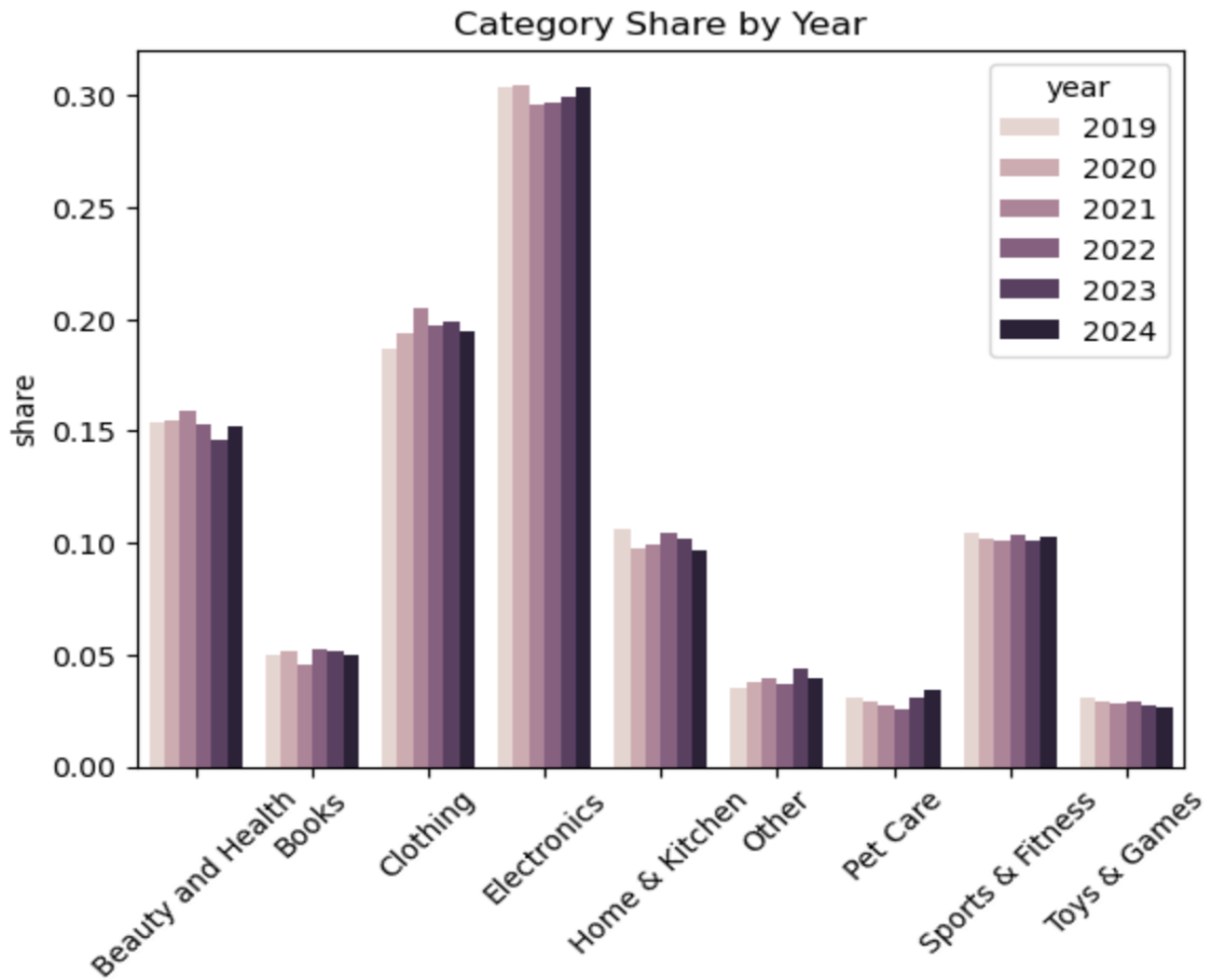
Key Insights

Main findings from the analysis:

- Revenue showed an overall positive growth trend over time.

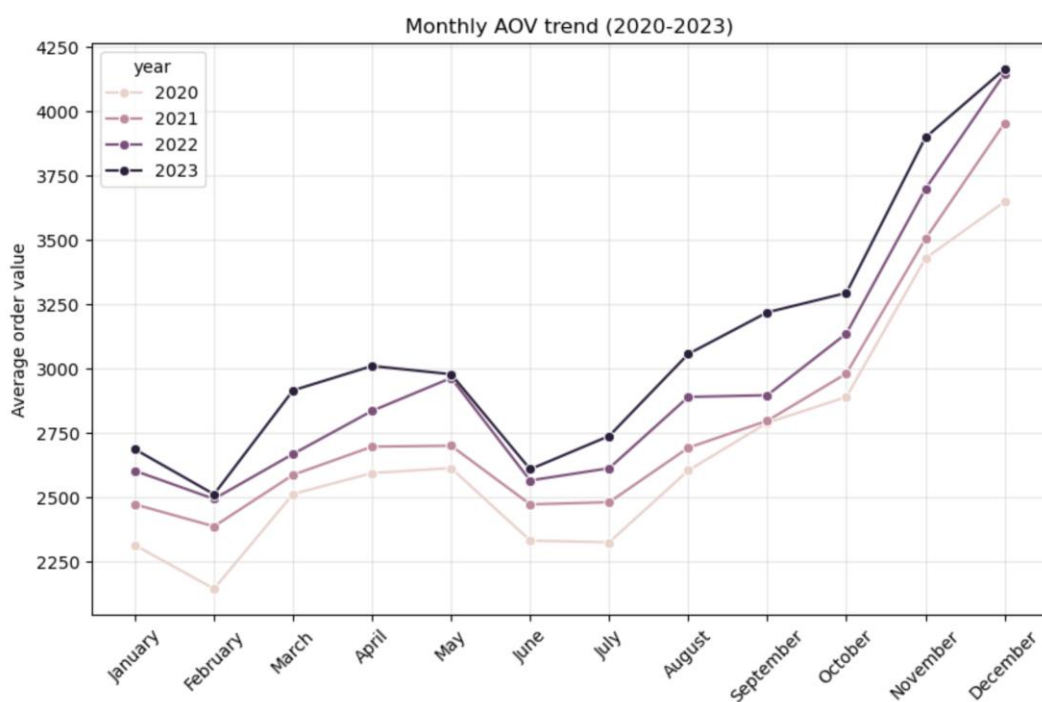


- Customers with repeated purchases generated a significant share of total revenue.
- Some product categories consistently outperformed others in sales volume and profitability.



- Discount did not significantly increase the number of orders, but reduced both average order value and revenue per user.
- This suggests that discounts are not effective and may be decreasing overall profitability.
- Key insights:
 - Discounts did not increase demand
 - The number of orders nearly identical.
 - Discounts are not driven additional purchases.
 - AOV decreased with discounts by 7%
 - ARPU decreased with discounts by 8%
 - Revenue is lower with discounts by 10%
 - The Customers were already 'ready to buy'
 - Discounts are not effective in this case.
- A relatively small group of loyal customers contributed a large percentage of total revenue.
- Retention metrics indicated opportunities for improving long-term customer engagement.

- In the recent period, the At Risk segment generated the highest share of revenue, indicating a growing risk of customer churn.
- Revenue contribution from Best Customers decreased compared to the historical RFM analysis.
- Loyal Customers still remain an important revenue driver, but their dominance weakened over time.
- Big Spenders became more visible in the recent analysis, suggesting that some customers generate high revenue despite lower purchase frequency.
- The increase in At Risk customers may indicate weakening customer retention and declining engagement among previously active users.
- Seasonal trends were visible in monthly sales dynamics.



Recommendations

Based on the analysis, the following recommendations were identified:

- Focus retention strategies on high-value customer segments.
- Improve loyalty programs to increase repeat purchases.
- Optimize discount campaigns to balance revenue growth and profitability.
- Invest more in high-performing product categories.
- Use customer segmentation for personalized marketing campaigns.
- Monitor retention and cohort metrics regularly to identify customer churn early.
- Limit blanket discounts
- Avoid giving discounts to all users

- Target discounts strategically
- Run A/B tests
- Compare discount vs no-discount groups
- Analyze customer segments
- Identify where discounts actually work

Technologies

The following tools and technologies were used in this project:

Programming & Analysis

Python:

- pandas
- matplotlib
- seaborn

Database & Querying

- SQL
- MySQL
- SQLAlchemy

Visualization & BI

- Power BI

Development Tools

- Jupyter Notebook
- Git
- GitHub

https://github.com/marinamaer123/Potfolio_ecommerce_project